explained the biosphere’s close interaction with the planet’s rock (lithosphere), water (hydrosphere) and air (atmosphere). This in turn led American biologist Eugene Odum to advocate a holistic approach to ecology. Odum argued that it was not possible to understand a single organism, or a group of organisms, without studying the ecosystem in which they live. He described this view as “the new ecology.”

In 1974, British scientist James Lovelock advanced the Gaia hypothesis that the interaction of living and nonliving elements in the biosphere reveal Earth to be a complex, self-regulating system that perpetuates the conditions for life. Almost two centuries earlier Hutton had articulated a similar idea—that biological and geological processes are interlinked and that Earth could be viewed as a superorganism. In Hutton’s words, “The globe of this earth is not just a machine but also an organized body as it has a regenerative power.”

Heading for extinction?

Life has survived on Earth for billions of years, despite the ravages of five mass extinctions. However, environmentalists now question whether it will survive another. Indeed, some contend that a sixth mass extinction has already started, as a result of human activity. Yet, if Lovelock’s Gaia theory is correct, it seems likely that the planet will endure—even if humans and many other current life forms do not.